

Berechtigungen in MySQL

DB erstellen

```
ddev config
```

vim .ddev/config.yaml

In diesem File die Datenbank und Version von **mariadb** -> **mysql** und **11.8** -> **8.4**

zur Datenbank “verbinden”

```
ddev mysql
```

alle User und deren Passwörter erstellen

```
CREATE USER 'db_admin'@'127.0.0.1' IDENTIFIED BY 'Einfach123!';  
CREATE USER 'db_admin'@'%' IDENTIFIED BY 'Einfach123!';
```

```
CREATE USER 'hr_mitarbeiter'@'127.0.0.1' IDENTIFIED BY 'Einfach123!';  
CREATE USER 'hr_mitarbeiter'@'%' IDENTIFIED BY 'Einfach123!';
```

```
CREATE USER 'praktikant'@'127.0.0.1' IDENTIFIED BY 'Einfach123!';  
CREATE USER 'praktikant'@'%' IDENTIFIED BY 'Einfach123!';
```

```
FLUSH PRIVILEGES;
```

```
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$ ddev mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 100
Server version: 8.4.5 Source distribution

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT user, host FROM mysql.user;
+-----+-----+
| user          | host          |
+-----+-----+
| db            | %            |
| db_admin      | %            |
| hr_mitarbeiter | %            |
| praktikant    | %            |
| root         | %            |
| db_admin      | 127.0.0.1    |
| hr_mitarbeiter | 127.0.0.1    |
| praktikant    | 127.0.0.1    |
| db            | localhost    |
| mysql.infoschema | localhost    |
| mysql.session | localhost    |
| mysql.sys     | localhost    |
| root         | localhost    |
+-----+-----+
13 rows in set (0.00 sec)

mysql> exit
Bye
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$
```

Berechtigungen vergeben

```
GRANT ALL PRIVILEGES ON schulverwaltung.* TO 'db_admin'@'127.0.0.1';
GRANT ALL PRIVILEGES ON schulverwaltung.* TO 'db_admin'@'%';
```

```
GRANT SELECT, INSERT, UPDATE, DELETE ON schulverwaltung.personen TO
'hr_mitarbeiter'@'127.0.0.1';
GRANT SELECT, INSERT, UPDATE, DELETE ON schulverwaltung.personen TO 'hr_mitarbeiter'@'%';
```

```
GRANT SELECT ON schulverwaltung.personen TO 'praktikant'@'127.0.0.1';
GRANT SELECT ON schulverwaltung.personen TO 'praktikant'@'%';
```

```
FLUSH PRIVILEGES;
```

```
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$ ddev mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 103
Server version: 8.4.5 Source distribution

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT user, host, Select_priv, Insert_priv, Update_priv, Delete_priv, Create_priv, Drop_priv
-> FROM mysql.user;
+-----+-----+-----+-----+-----+-----+-----+-----+
| user          | host      | Select_priv | Insert_priv | Update_priv | Delete_priv | Create_priv | Drop_priv |
+-----+-----+-----+-----+-----+-----+-----+-----+
| db            | %         | N           | N           | N           | N           | N           | N         |
| db_admin      | %         | N           | N           | N           | N           | N           | N         |
| hr_mitarbeiter | %         | N           | N           | N           | N           | N           | N         |
| praktikant    | %         | N           | N           | N           | N           | N           | N         |
| root          | %         | Y           | Y           | Y           | Y           | Y           | Y         |
| db_admin      | 127.0.0.1 | N           | N           | N           | N           | N           | N         |
| hr_mitarbeiter | 127.0.0.1 | N           | N           | N           | N           | N           | N         |
| praktikant     | 127.0.0.1 | N           | N           | N           | N           | N           | N         |
| db            | localhost | N           | N           | N           | N           | N           | N         |
| mysql.infoschema | localhost | Y           | N           | N           | N           | N           | N         |
| mysql.session | localhost | N           | N           | N           | N           | N           | N         |
| mysql.sys     | localhost | N           | N           | N           | N           | N           | N         |
| root          | localhost | Y           | Y           | Y           | Y           | Y           | Y         |
+-----+-----+-----+-----+-----+-----+-----+-----+
13 rows in set (0.00 sec)

mysql> exit
Bye
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$
```

Berechtigungen testen

```
USE schulverwaltung;
```

db_admin

```
mysql -h 127.0.0.1 -P 32772 -u db_admin -p
```

Test A

```
SELECT * FROM personen;
```

Test B

```
ALTER TABLE personen ADD COLUMN test_info VARCHAR(255);
ALTER TABLE personen DROP COLUMN test_info;
```

```
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$ mysql -h 127.0.0.1 -P 32772 -u db_admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 82
Server version: 8.4.5 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE schulverwaltung;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [schulverwaltung]> SELECT * FROM personen;
+----+-----+-----+-----+-----+
| id | vorname | nachname | svnr | geb_datum |
+----+-----+-----+-----+-----+
| 1 | Anna | Müller | 1234010105 | 2005-01-01 |
| 2 | Lukas | Wagner | 5678150306 | 2006-03-15 |
| 3 | Sarah | Huber | 9012200705 | 2005-07-20 |
| 4 | Max | Mustermann | 1111101010 | 2010-10-10 |
+----+-----+-----+-----+-----+
4 rows in set (0.000 sec)

MySQL [schulverwaltung]> ALTER TABLE personen ADD COLUMN test_info VARCHAR(255);
Query OK, 0 rows affected (0.055 sec)
Records: 0 Duplicates: 0 Warnings: 0

MySQL [schulverwaltung]> ALTER TABLE personen DROP COLUMN test_info;
Query OK, 0 rows affected (0.009 sec)
Records: 0 Duplicates: 0 Warnings: 0

MySQL [schulverwaltung]> exit
Bye
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$
```

hr_mitarbeiter

```
mysql -h 127.0.0.1 -P 32772 -u hr_mitarbeiter -p
```

Test A

```
UPDATE personen SET nachname = 'Müller-Hof' WHERE id = 1;
```

Test B

```
INSERT INTO personen (vorname, nachname, svnr, geb_datum)
VALUES ('Felix', 'Test', '9999010199', '1999-01-01');
```

Test C

```
SELECT * FROM personen_log;
```

Test D

```
DROP TABLE personen;
```

```
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$ mysql -h 127.0.0.1 -P 32772 -u hr_mitarbeiter -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 89
Server version: 8.4.5 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE schulverwaltung;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [schulverwaltung]> UPDATE personen SET nachname = 'Müller-Hof' WHERE id = 1;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MySQL [schulverwaltung]> INSERT INTO personen (vorname, nachname, svnr, geb_datum)
-> VALUES ('Felix', 'Test', '9999010199', '1999-01-01');
Query OK, 1 row affected (0.001 sec)

MySQL [schulverwaltung]> SELECT * FROM personen_log;
ERROR 1142 (42000): SELECT command denied to user 'hr_mitarbeiter'@'172.20.0.1' for table 'personen_log'
MySQL [schulverwaltung]> DROP TABLE personen;
ERROR 1142 (42000): DROP command denied to user 'hr_mitarbeiter'@'172.20.0.1' for table 'personen'
MySQL [schulverwaltung]> exit
Bye
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$
```

praktikant

Test A

```
SELECT * FROM personen;
```

Test B

```
UPDATE personen SET vorname = 'Böser Bube' WHERE id = 1;
```

Test C

```
INSERT INTO personen (vorname, nachname, svnr, geb_datum)
VALUES ('Hacker', 'Mensch', '0000010100', '2000-01-01');
```

Test D

```
SELECT * FROM personen_log;
```

```

fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$ mysql -h 127.0.0.1 -P 32772 -u praktikant -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 98
Server version: 8.4.5 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE schulverwaltung;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [schulverwaltung]> SELECT * FROM personen;
+-----+-----+-----+-----+-----+
| id | vorname | nachname | svnr | geb_datum |
+-----+-----+-----+-----+-----+
| 1 | Anna | Müller-Hof | 1234010105 | 2005-01-01 |
| 2 | Lukas | Wagner | 5678150306 | 2006-03-15 |
| 3 | Sarah | Huber | 9012200705 | 2005-07-20 |
| 4 | Max | Mustermann | 1111101010 | 2010-10-10 |
| 9 | Felix | Test | 9999010199 | 1999-01-01 |
+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MySQL [schulverwaltung]> UPDATE personen SET vorname = 'Böser Bube' WHERE id = 1;
ERROR 1142 (42000): UPDATE command denied to user 'praktikant'@'172.20.0.1' for table 'personen'
MySQL [schulverwaltung]> INSERT INTO personen (vorname, nachname, svnr, geb_datum)
-> VALUES ('Hacker', 'Mensch', '0000010100', '2000-01-01');
ERROR 1142 (42000): INSERT command denied to user 'praktikant'@'172.20.0.1' for table 'personen'
MySQL [schulverwaltung]> SELECT * FROM personen_log;
ERROR 1142 (42000): SELECT command denied to user 'praktikant'@'172.20.0.1' for table 'personen_log'
MySQL [schulverwaltung]> exit
Bye
fritzi@Bleach:~/Desktop/SCHULE/3BHITS/INSY/Berechtigungen_MySQL$

```